



Written exam
„Numerical Optimization and Model Predictive Control“
on

Please fill out legibly:

Last name, first name:

Student registration number:

I hereby declare that I am eligible for examination and have read the instructions on page 3.

Signature:

Total number of printed sheets: 12

Good luck!

Task	1	2	3	4	Sum	Grade
Points	/ 24	/ 26	/ 20	/ 20	/ 90	

Information about the exam

Evaluation of the exam

- If you are not on the list of registered students, your participation at the exam is considered with reservation.
- Results without comprehensible argumentation or without clear mathematical derivation cannot be accounted for.
- Use only document-proof pens. Do not use red color.

Notes on the exam

- Subtasks marked with * are new tasks (no follow-up), whereas subtasks without * build upon previous ones.
- Answer the subtasks in the prepared areas. Use the enclosed spare pages if there is not enough space. In any case, mark all subtasks with the task number (e.g. „1c“).
- Please do not remove the staples.

Tools

- The lecture notes, the exercise materials and transcripts can be used in the exam. Also computer-written notes, summaries and formularies with clear reference to the course may be used in printed form.
- Previous exams, books, calculators and materials from other courses are not permitted.

Task 1 (24 Points)

Given is the linear constrained optimization problem with parameter s

$$\min_{\mathbf{x} \in \mathbb{R}^2} f(\mathbf{x}, s) = (1 - s)x_1 - x_2 \quad (1a)$$

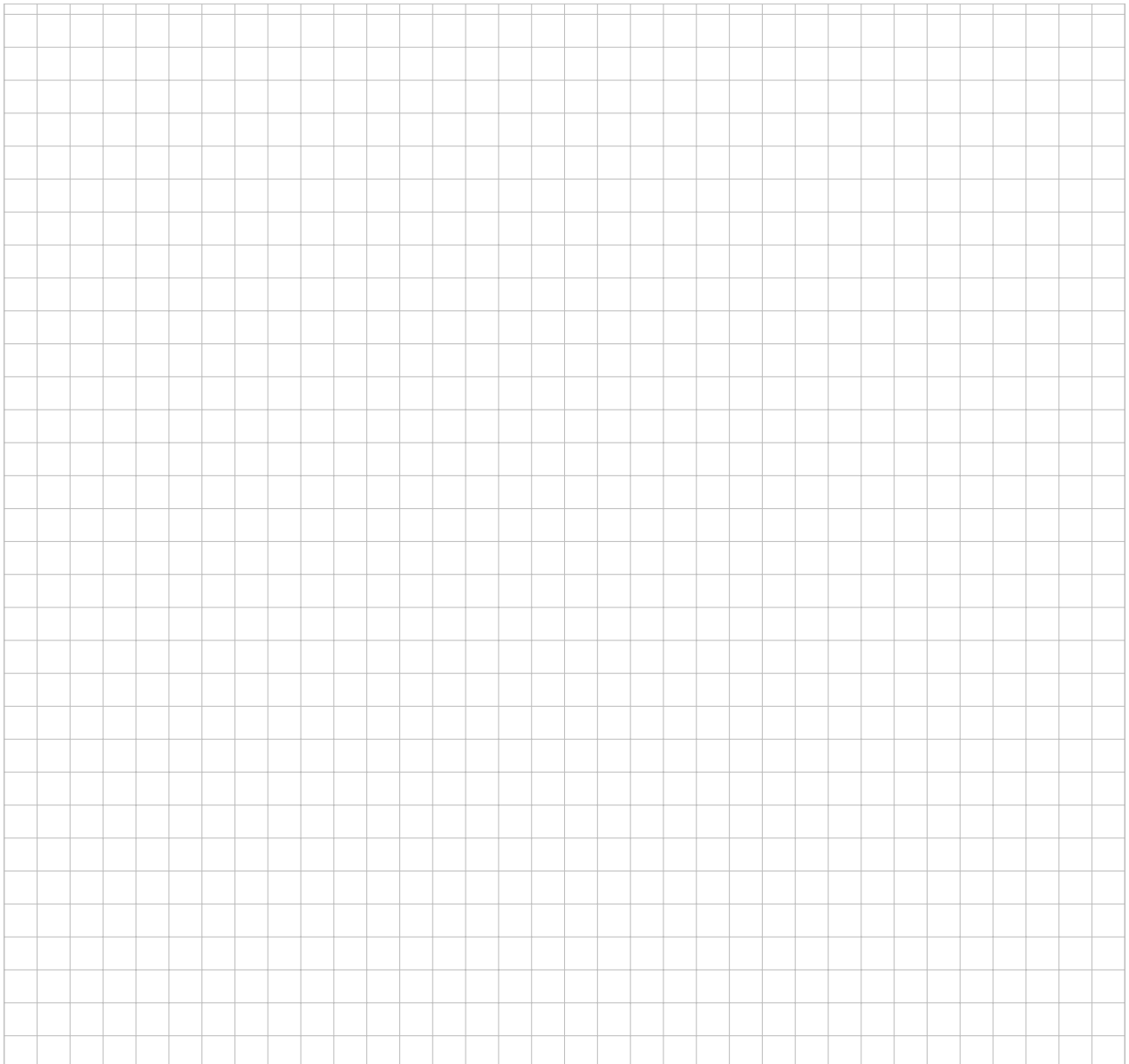
$$\text{s.t. } h_1(\mathbf{x}) = 2x_1 + x_2 - 4 \leq 0 \quad (1b)$$

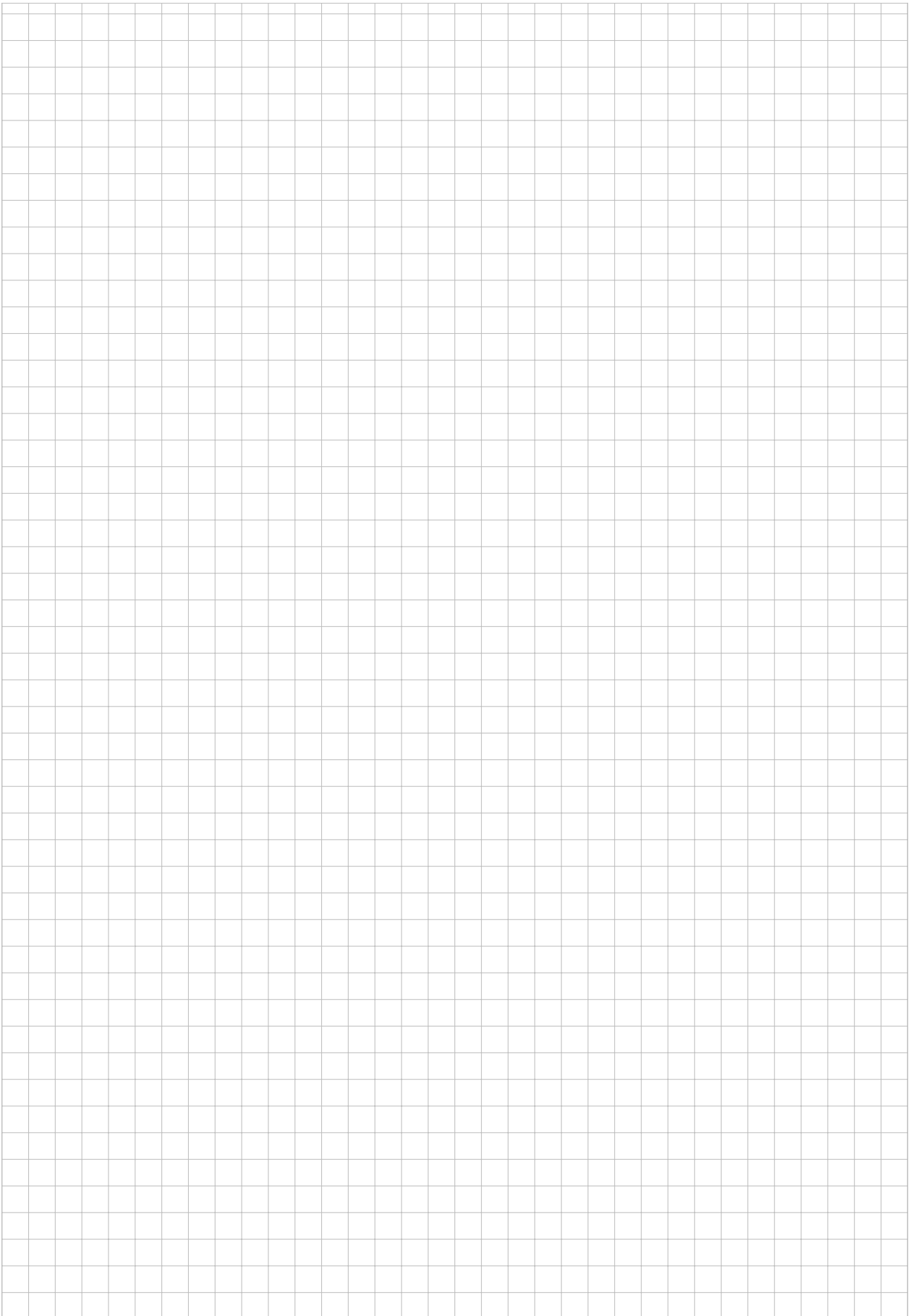
$$h_2(\mathbf{x}) = -x_1 \leq 0 \quad (1c)$$

$$h_3(\mathbf{x}) = -x_2 \leq 0. \quad (1d)$$

(2 P) *a) Formulate the Lagrangian function and the first-order optimality conditions for problem (1) without bringing it to standard form.

(10 P) b) Determine the optimal solution $\mathbf{x}^*$ in dependence of s on the interval $0 \leq s \leq 4$.
Hint: Use the optimality conditions from Task 1a) and consider the different cases of active constraints to divide the interval $0 \leq s \leq 4$ into sections.





Now s is set to $\frac{1}{2}$ and the inequality constraints (1b) and (1d) are replaced by new constraints, resulting in the optimization problem

$$\min_{\mathbf{x} \in \mathbb{R}^2} \quad \frac{1}{2}x_1 - x_2 \quad (2a)$$

$$\text{s.t.} \quad x_1 \geq 0 \quad (2b)$$

$$x_2 \leq 2. \quad (2c)$$

In the following, interior penalty functions (also called barrier functions) that penalize the constraints (2b), (2c) shall be used to solve the problem.

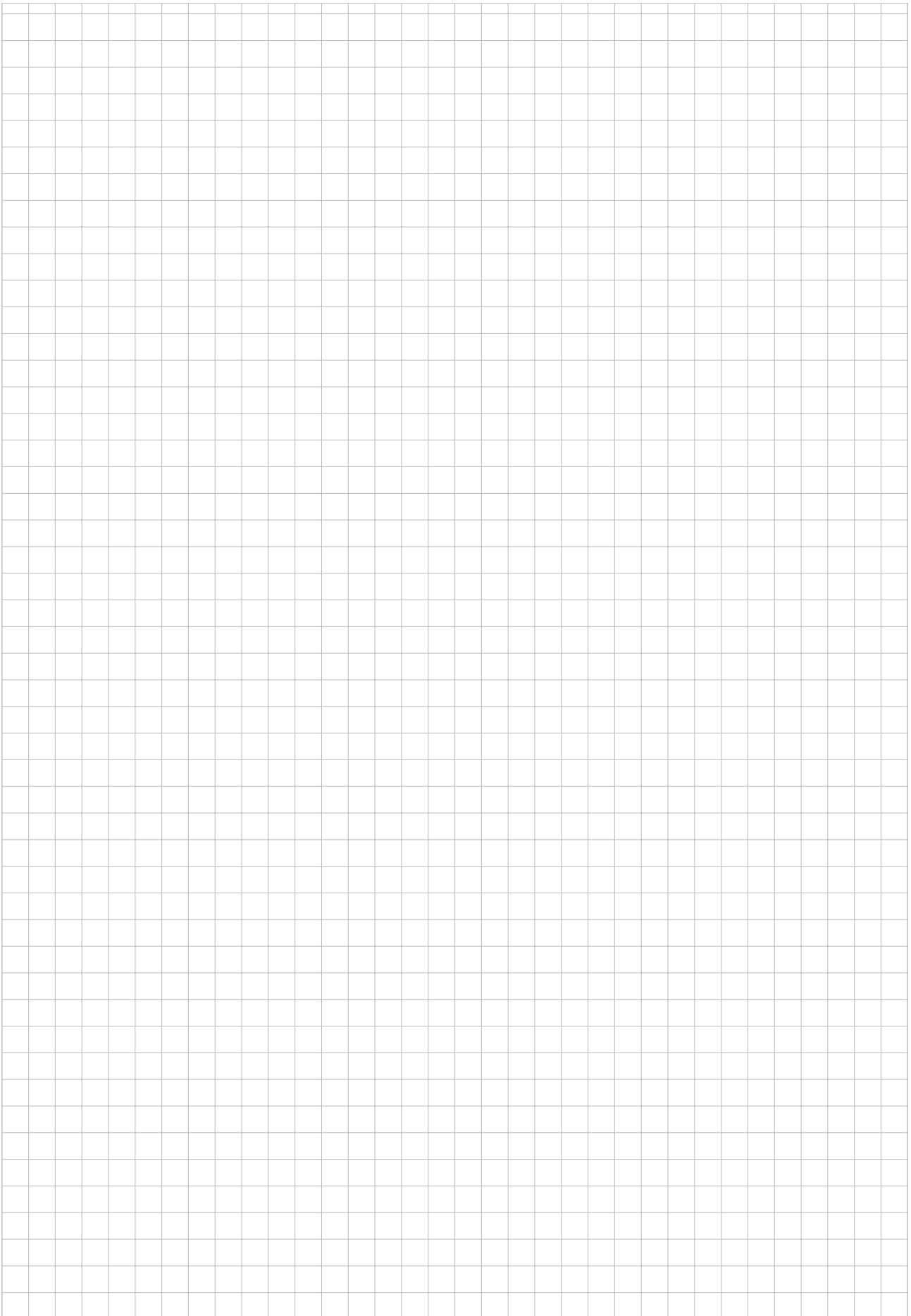
(6 P) * c) Formulate the resulting unconstrained problem with penalty parameter $\tau > 0$ when using the interior penalty function $\psi(h(\mathbf{x})) = -\ln(-h(\mathbf{x}))$ for $h(\mathbf{x}) \leq 0$. Then calculate the optimal solution $\mathbf{x}^*(\tau)$ of the resulting problem as a function of τ . How does τ influence the solution?

(6 P) d) Perform one step of the gradient method to solve the problem from Task 2c). Start with

$$\mathbf{x}^0 = \left[1, \frac{3}{2} \right]^T, \quad \tau = \frac{1}{4}$$

and choose the step size $\alpha^0 > 0$ as large as possible and justify your choice.





Task 2 (26 Points)

Given is the quadratic constrained optimization problem

$$\min_{\mathbf{x} \in \mathbb{R}^2} f(\mathbf{x}) = 2(x_1 - 1.5)^2 + (x_2 - 2.5)^2 \quad (3a)$$

$$\text{s.t. } h_1(\mathbf{x}) = 2x_1 + x_2 - 3.5 \leq 0 \quad (3b)$$

$$h_2(\mathbf{x}) = -0.5x_1 + x_2 - 1 \leq 0 \quad (3c)$$

$$h_3(\mathbf{x}) = x_1 \geq 0 \quad (3d)$$

$$h_4(\mathbf{x}) = x_2 \geq 0. \quad (3e)$$

(5 P) * a) Draw the constraints and the admissible set into Figure 1. Is the problem convex?

(4 P) * b) Formulate the Lagrangian function and the first-order optimality conditions for problem (3).

(10 P) c) Solve problem (3) with the active set method starting from the point $\mathbf{x}^0 = [0.5, 0]^T$. Initialize the working set $\mathcal{A}^0$ with the constraints being active at the starting point. At each iteration step, graphically determine the solution to the equality-constrained quadratic problem of the current working set using the contour lines and admissible set in Figure 1. Then compute the Lagrangian multipliers to determine the new working set.

(2 P) d) Discuss the meaning of the resulting Lagrangian multipliers at the optimal point.



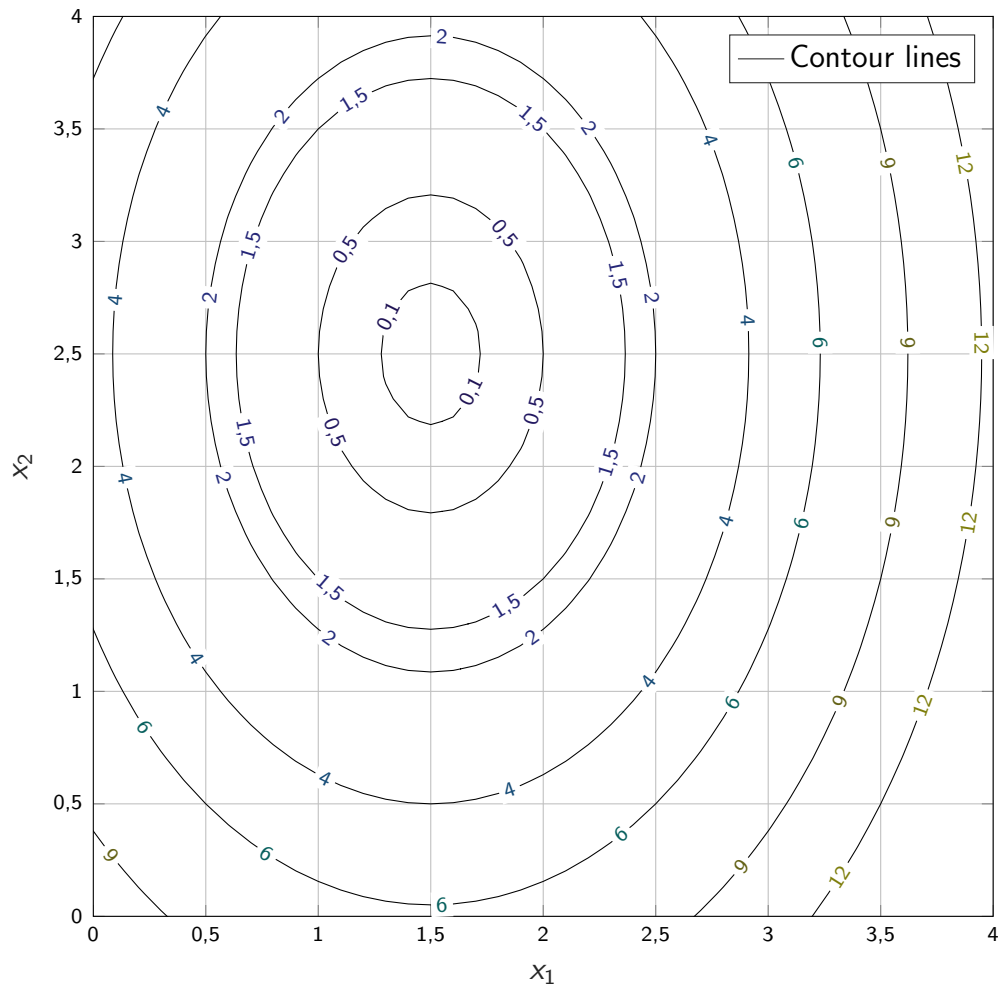
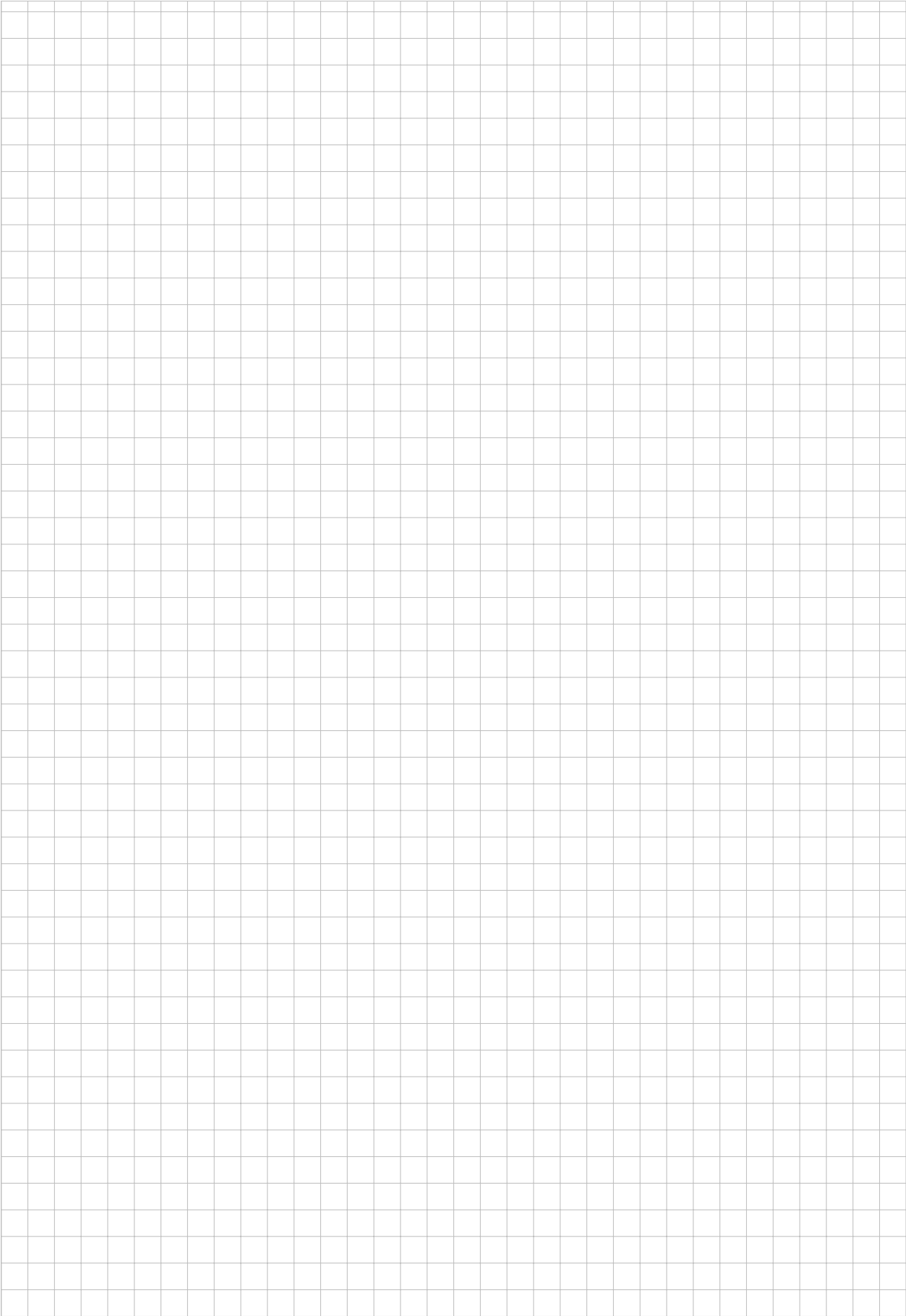


Figure 1: Coordinate system for Task 2.

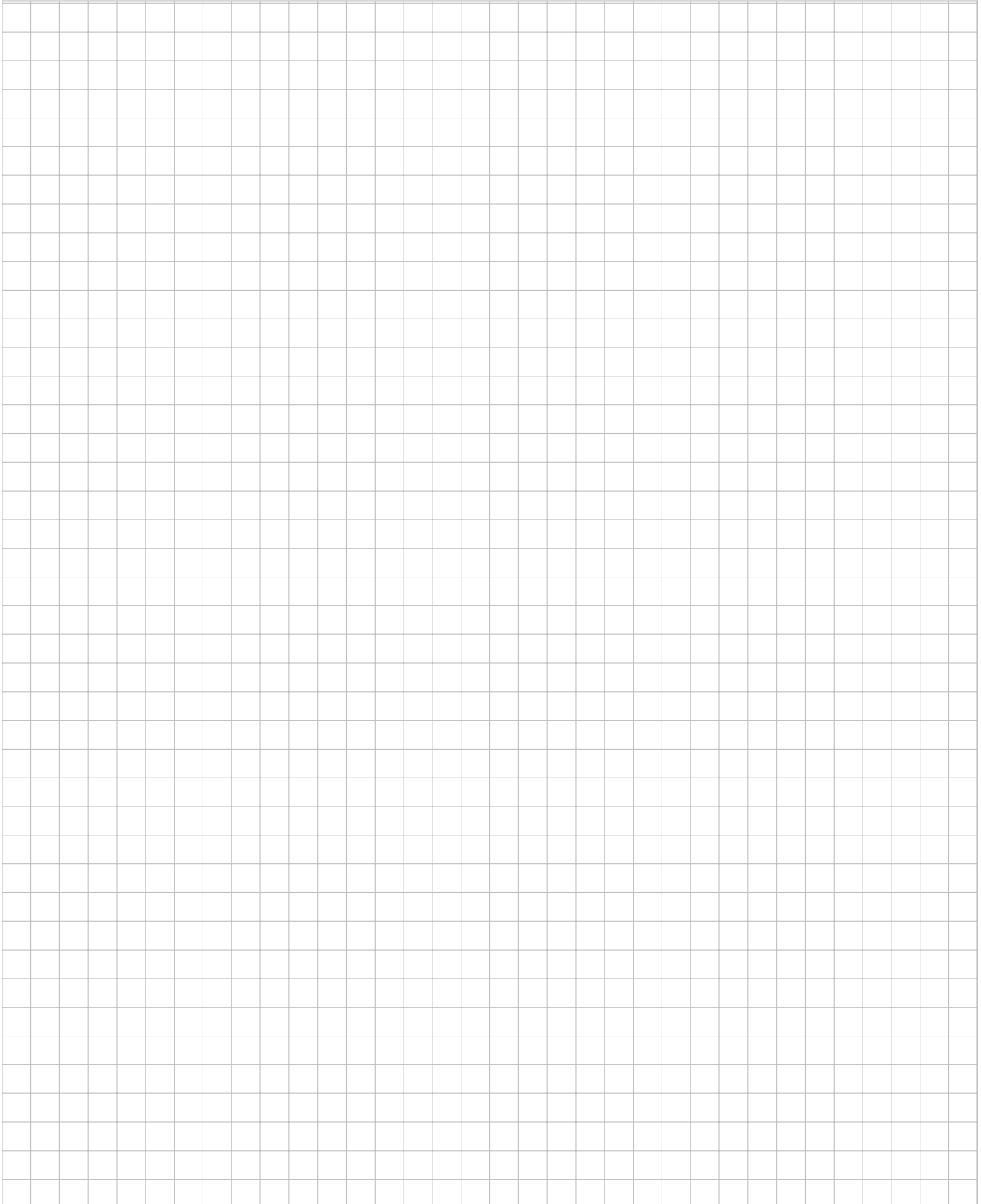




In the following, the constraint $h_2(\mathbf{x}) \leq 0$ in (3c) is replaced by the parabolic constraint

$$\hat{h}_2(\mathbf{x}) = 8x_1^2 + 0.5 - x_2 \leq 0. \quad (4)$$

- * e) Draw the constraint (4) into Figure 1 and determine the new optimal solution x^* graphically. (5 P)
Is the new problem convex? Would the active set method be suitable to solve this problem?



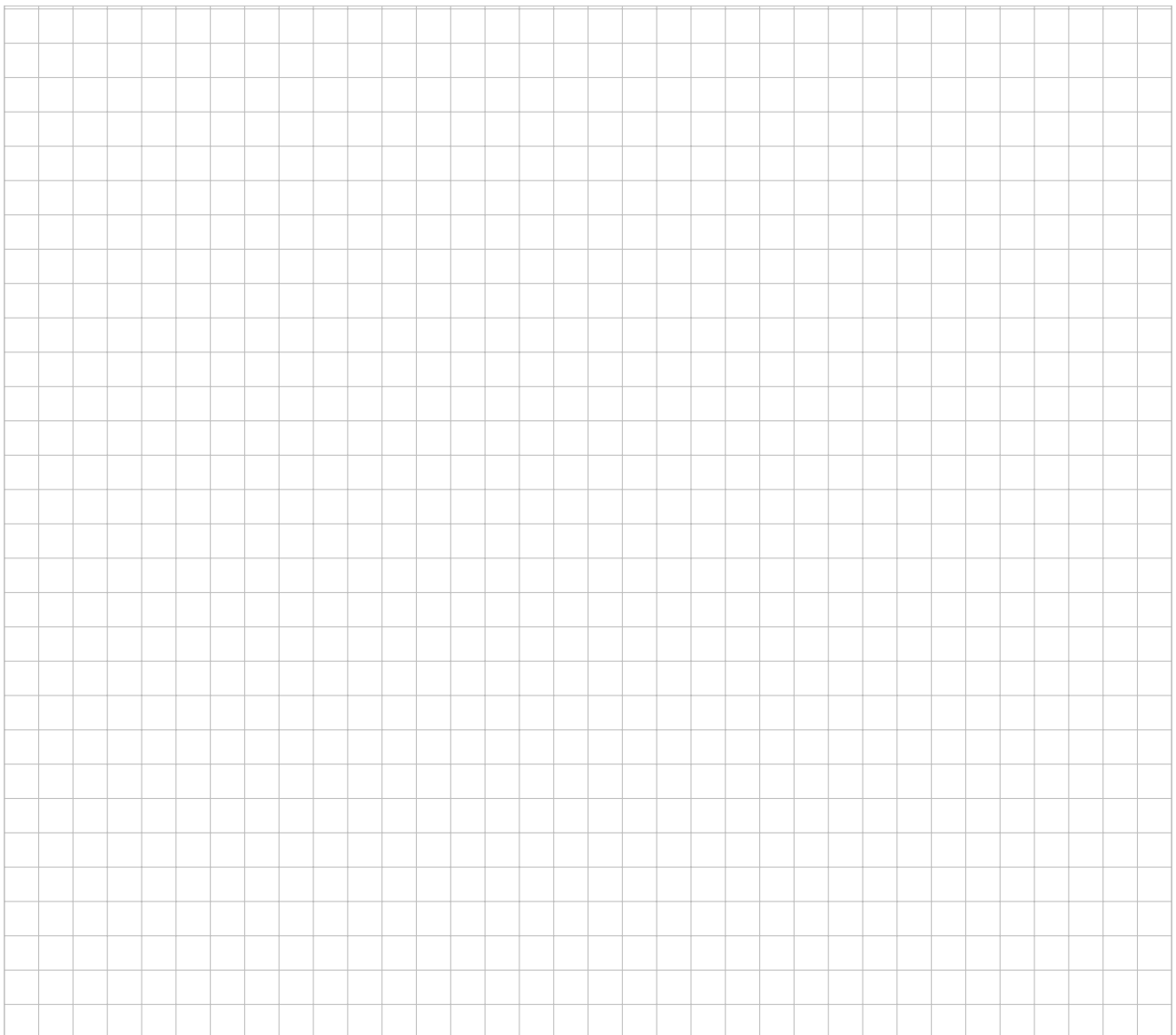
Task 3 (20 Points)

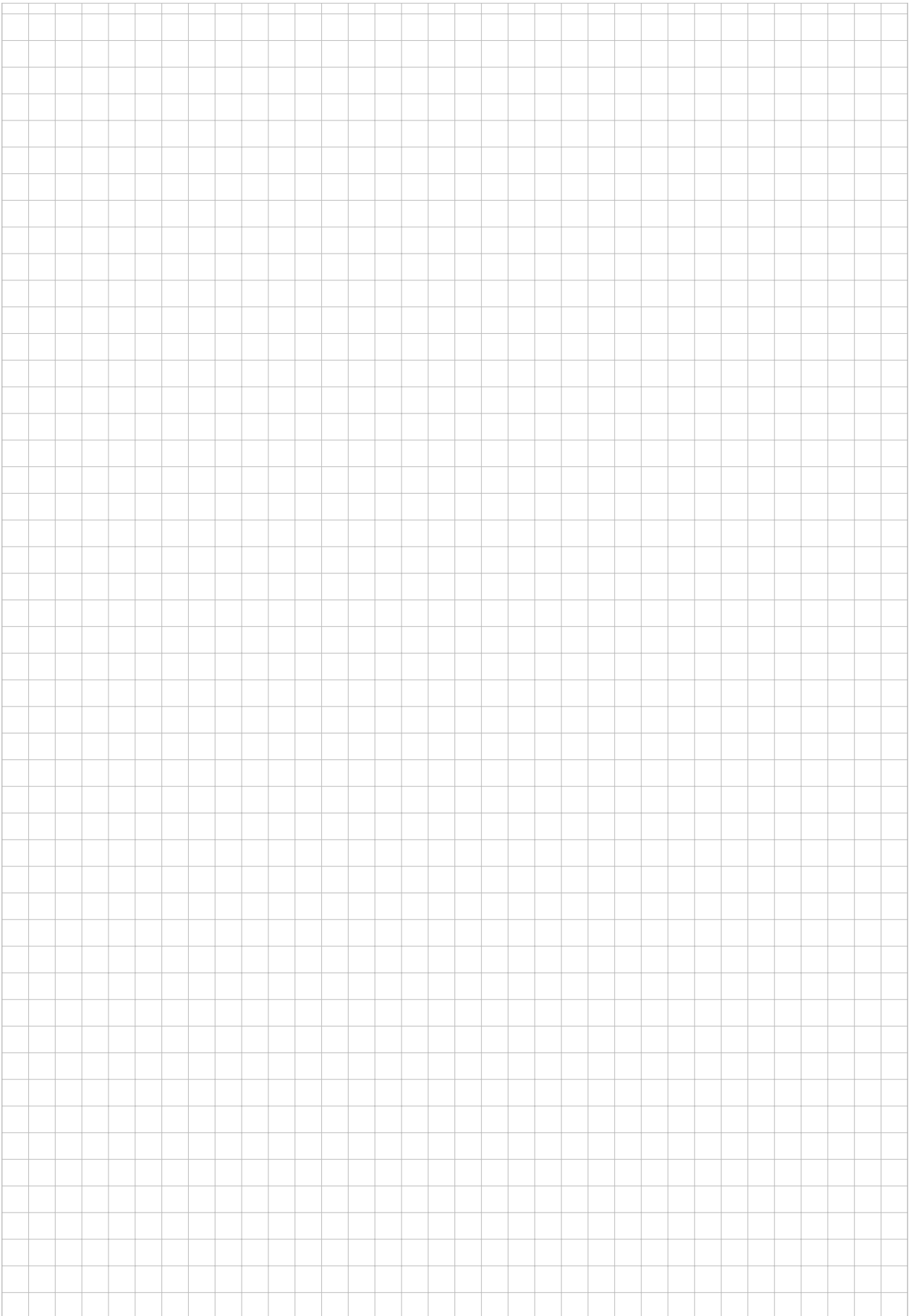
Consider the following dynamical optimization problem with parameter $a > 0$ and free end time t_f

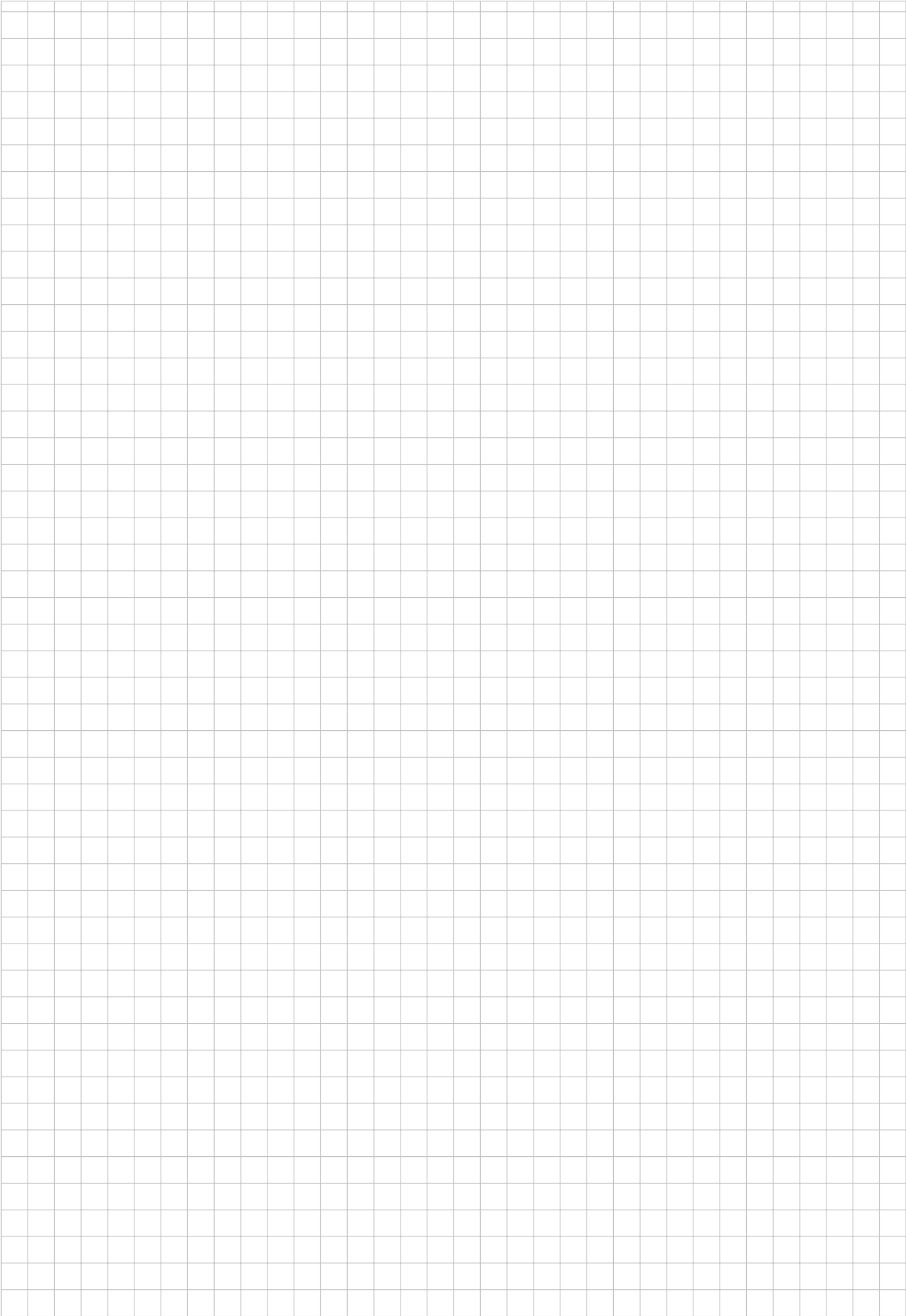
$$\min_{u(\cdot)} \int_0^{t_f} ax + \frac{1}{2}u^2 dt \quad (5a)$$

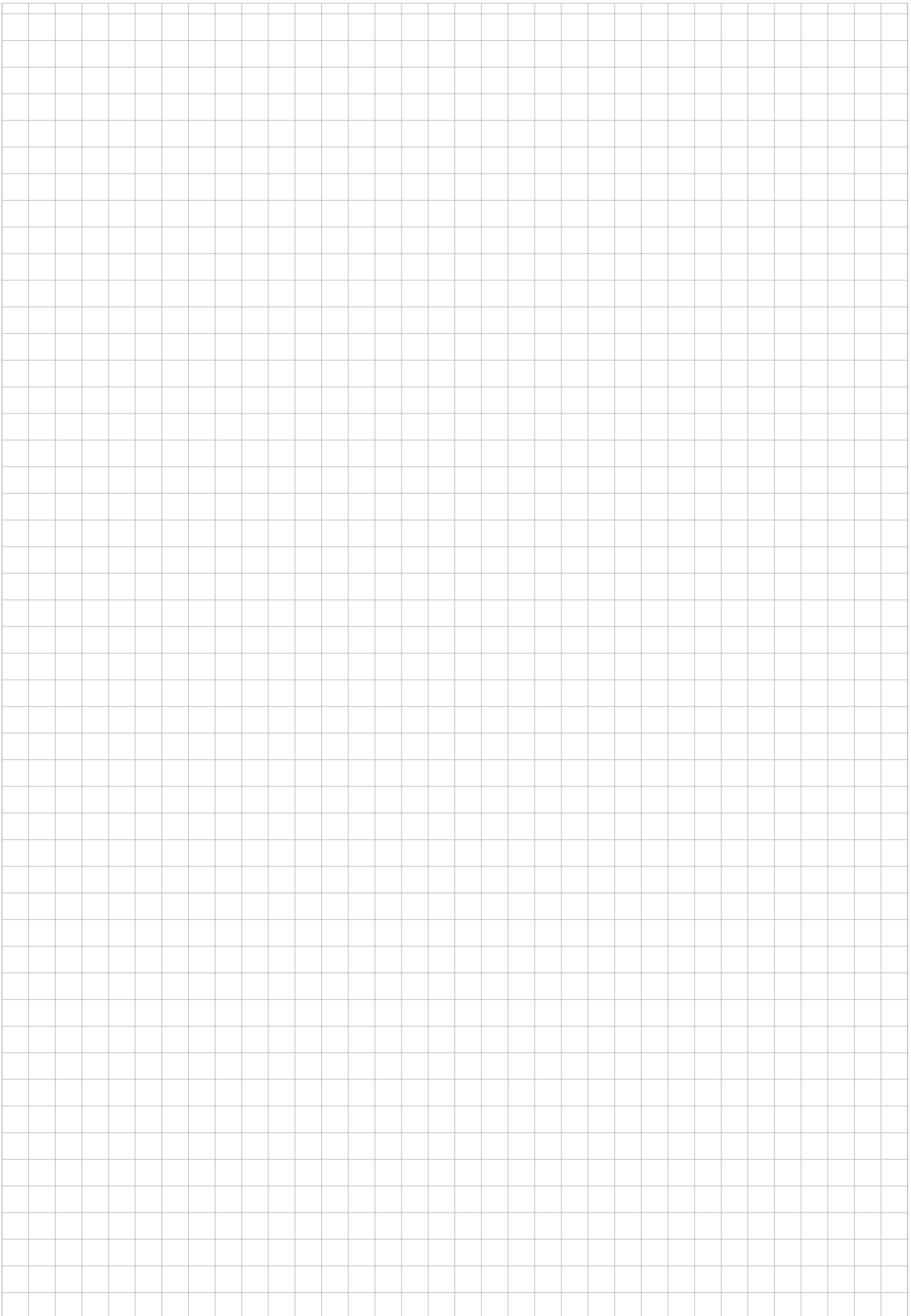
$$\text{s.t. } \dot{x} = u, \quad x(0) = 1, \quad x(t_f) = 0. \quad (5b)$$

- (4 P) *a) State the Hamiltonian function for problem (5) and derive the necessary first-order optimality conditions.
- (10 P) b) Calculate the optimal trajectories $x^*(t)$, $u^*(t)$ and $\lambda^*(t)$ as well as the optimal end time t_f^* for the given problem in dependence of parameter $a > 0$.
- (2 P) *c) Which effect does the parameter a have on the resulting optimal trajectories and the end time?
- (4 P) *d) What is the optimal control input $u^*(t)$ when an additional input constraint $u \in [-1, 1]$ is considered?









Task 4 (20 Points)

- (6 P) *a) The solution behavior of the augmented Lagrangian algorithm should be analyzed for the optimization problem

$$\min_{x \in \mathbb{R}} f(x) = (x - 2)^2 \quad (6a)$$

$$\text{s.t. } g(x) = 2x - 1 = 0. \quad (6b)$$

What is the optimal solution x^* and corresponding multiplier λ^* ? State the augmented Lagrangian function $L_c(x, \lambda)$ for (6) and calculate

$$x^k = \arg \min_x L_{c^k}(x, \lambda^k) \quad (7)$$

in dependence of c^k and λ^k .

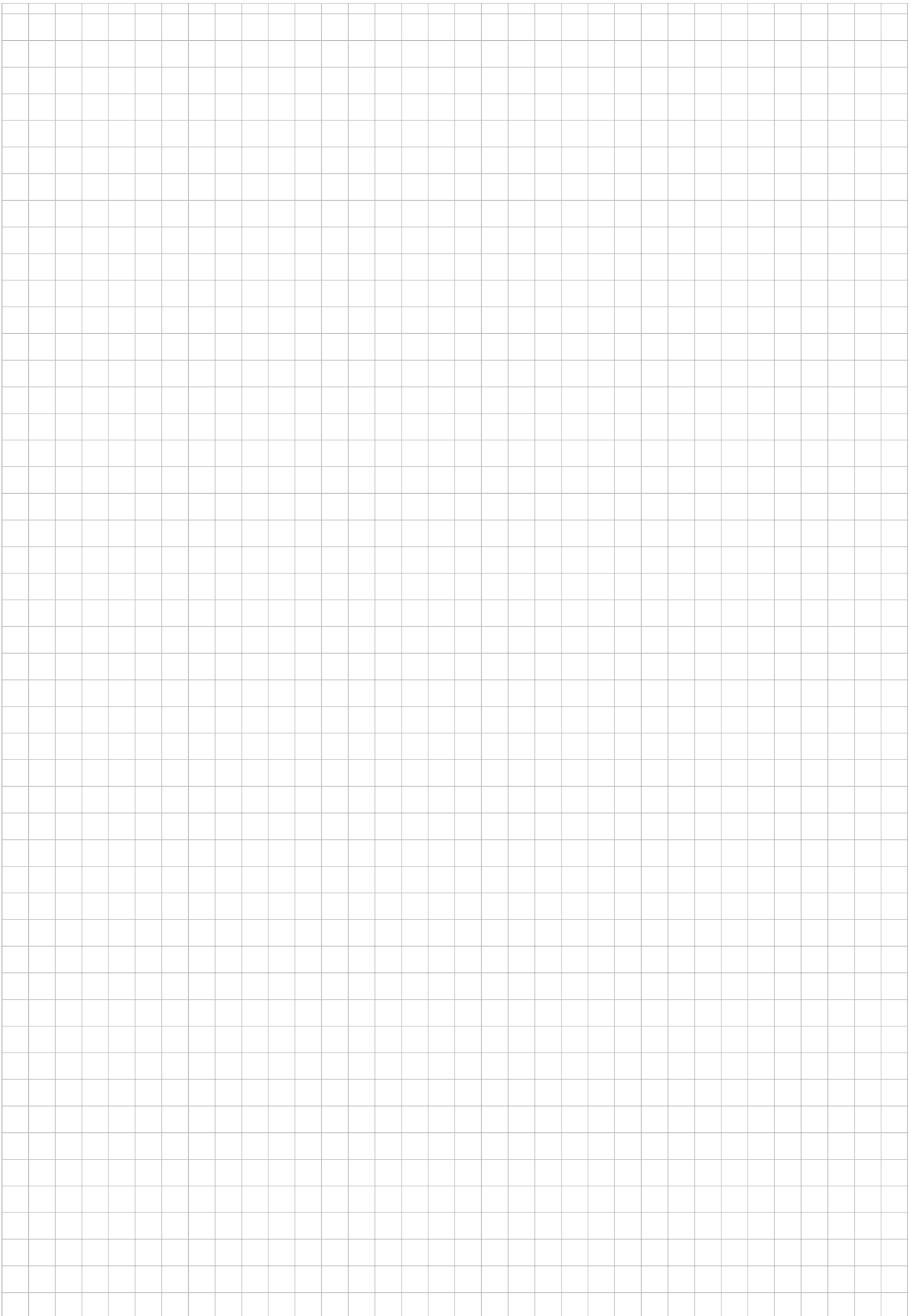
- (4 P) *b) Investigate the convergence of $\lambda^{k+1} = \lambda^k + c^k g(x^k)$ for problem (6) by calculating $C > 0$ for

$$|\lambda^{k+1} - \lambda^*| \leq C |\lambda^k - \lambda^*|. \quad (8)$$

For which values of the penalty parameter c^k does the augmented Lagrangian method lead to the optimal solution?

If Task 4a) could not be solved, assume $x^k = \frac{c^k - \lambda^k + 2}{2 + 2c^k}$ and $\lambda^* = 1$.





(6 P) * c) Consider the MPC problem

$$\min_{u(\cdot)} \quad p x(T)^2 + \int_0^T q x^2 + r u^2 dt \quad (9a)$$

$$\text{s.t.} \quad \dot{x} = ax + bu, \quad x(0) = x_0 \quad (9b)$$

$$x(T) = S_\beta \quad (9c)$$

with the positive constants $p, q, r > 0$. The system is assumed to be stabilizable and p is the solution of the algebraic Riccati equation

$$pa + ap - p^2 b^2 / r + q = 0. \quad (10)$$

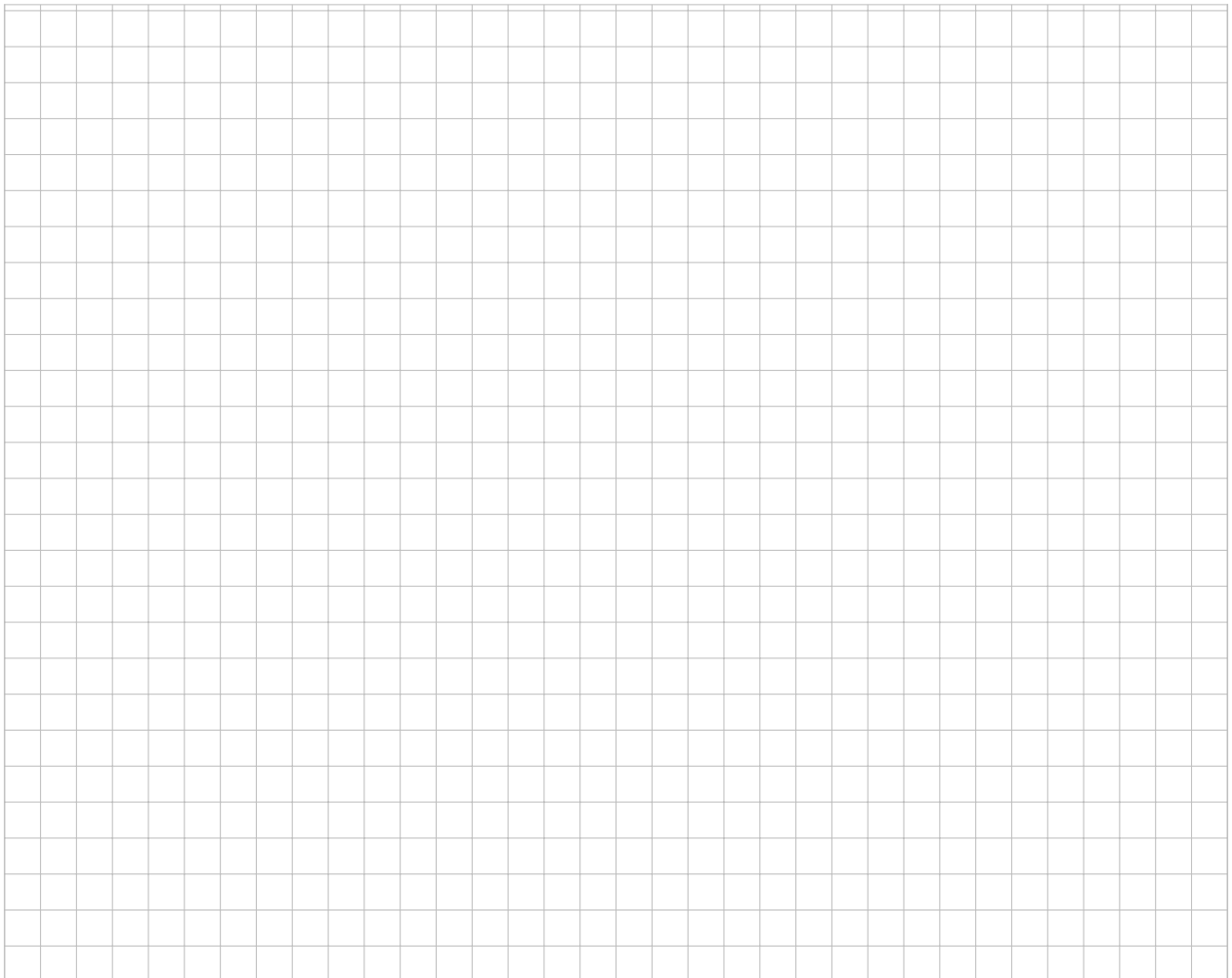
Show, that in this case the CLF inequality

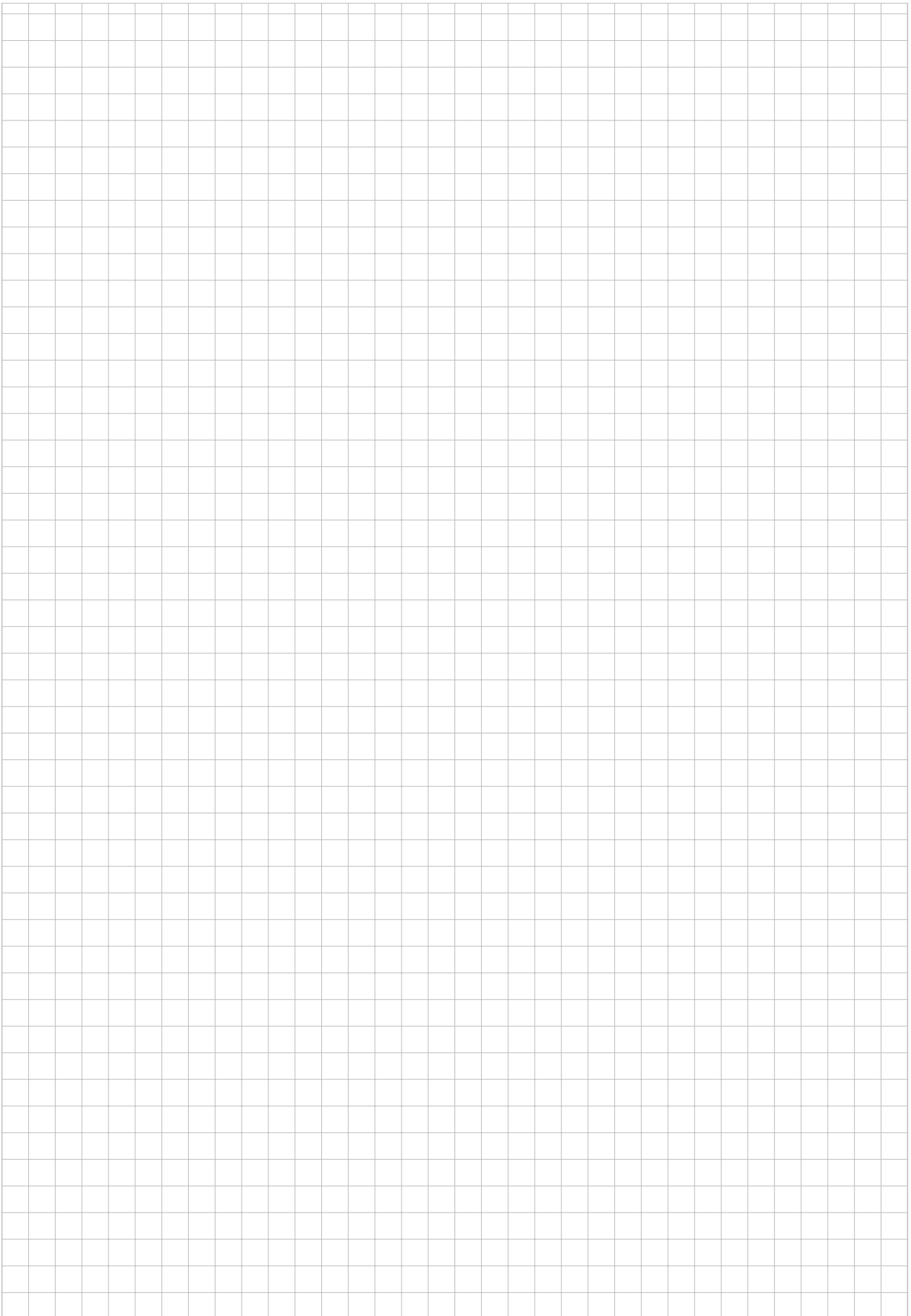
$$\dot{V}(x) + l(x, -kx) \leq 0, \quad \forall x \in S_\beta \quad (11)$$

is fulfilled with control law

$$u = -kx, \quad \text{where } k = bp/r. \quad (12)$$

Give the corresponding feasible terminal region S_β .





(4 P) *d) Given is the dynamical optimization problem

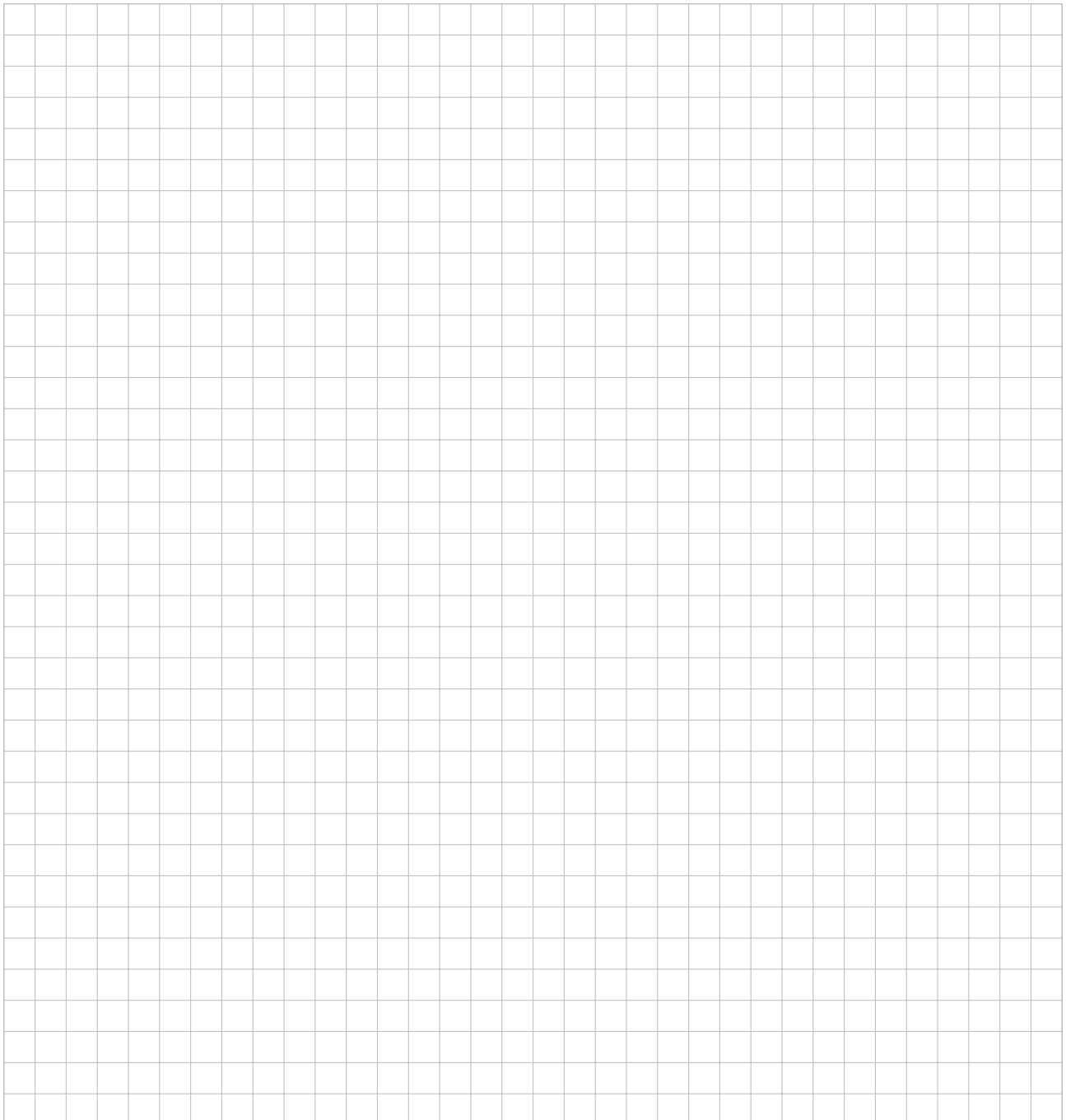
$$\min_{\mathbf{u}(\cdot)} V(\mathbf{x}(t_f), t_f) + \int_{t_0}^{t_f} l(\mathbf{x}(t), \mathbf{u}(t)) dt \quad (13a)$$

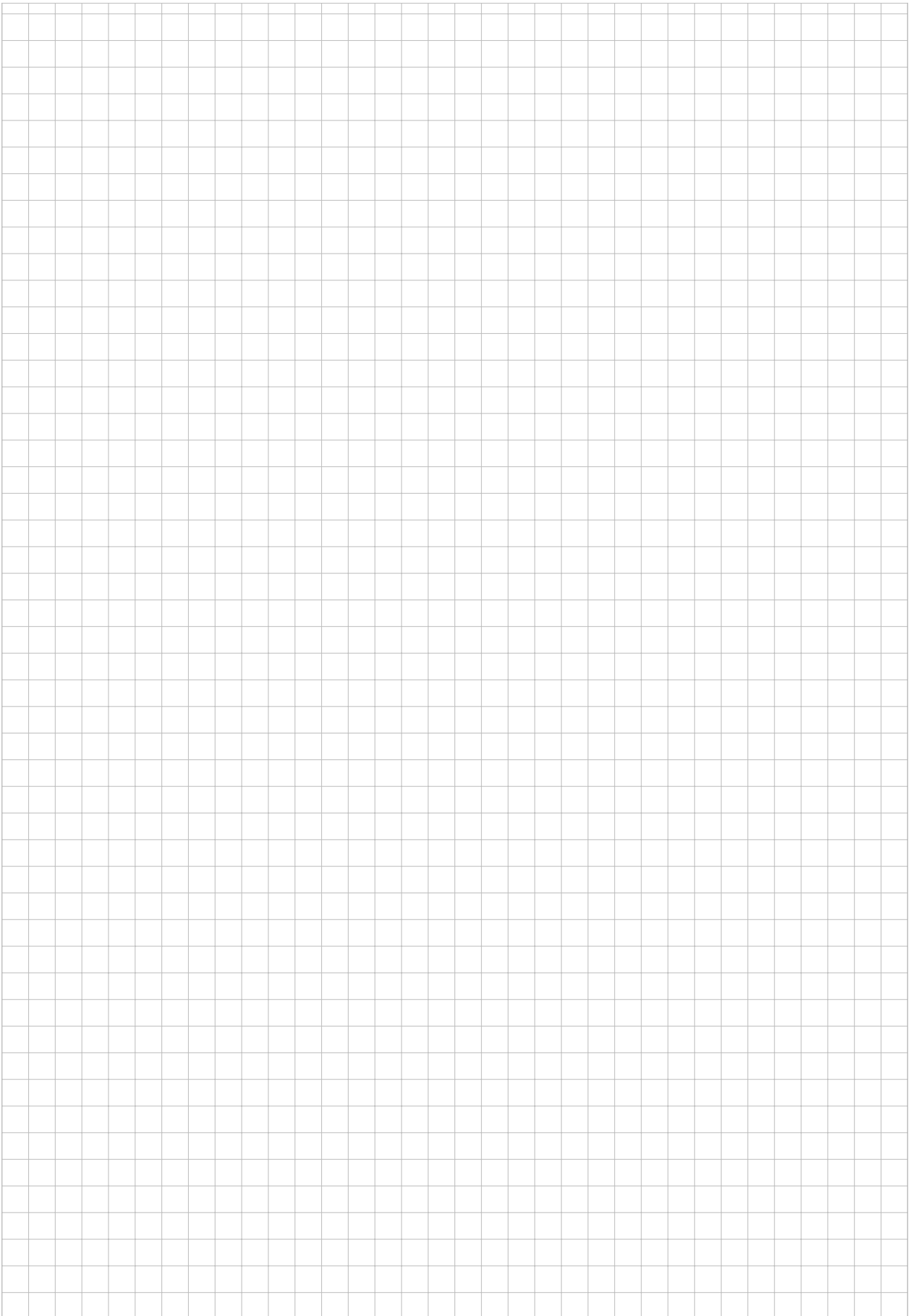
$$\text{s.t. } \dot{\mathbf{x}}(t) = f(\mathbf{x}(t), \mathbf{u}(t)) \quad (13b)$$

$$x_i(t_0) = x_{0,i}, \quad i \in \mathcal{I}_0 \quad (13c)$$

$$x_j(t_f) = x_{f,j}, \quad j \in \mathcal{I}_f \quad \text{with} \quad \mathcal{I}_0 \cap \mathcal{I}_f = \emptyset \quad (13d)$$

with n partial initial and terminal conditions for the state $\mathbf{x} \in \mathbb{R}^n$. State the boundary conditions from the first-order optimality conditions. Which additional boundary conditions would you suggest in this case? (Hint: A proof using the calculus of variations is not necessary).





Spare pages

